BVS1 · Winter Semester 2026/2027 · TH Köln · ICCT

Handout — Editing Source Files in Ubuntu — nano and Vim

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You only need one editor for BVS1. If you have not used a terminal editor before, **nano is usually the quickest way to get started**. Use **Vim** if you already know it or want to learn modal editing.

The editor is not part of the assessment. The goal is simply to edit source files reliably.

nano — quickest start

Open a file:

```
nano main.c
```

Nano immediately lets you type and edit text. At the bottom of the screen, `^` means **Ctrl**. For example, `^O` means **Ctrl-O**.

Action	Key
Save	**Ctrl-O**, then **Enter**
Exit	**Ctrl-X**
Search	**Ctrl-W**
Go to line	**Ctrl-_**
Cut current line	**Ctrl-K**
Paste last cut text	**Ctrl-U**
Undo	**Alt-U**
Redo	**Alt-E**
Help	**Ctrl-G**

A safe everyday sequence is:

```
nano main.c
    ↓
edit
    ↓
Ctrl-O, Enter
    ↓
Ctrl-X
```

If Nano asks whether to save modified content when you leave, answer deliberately. Do not press keys blindly.

Vim — minimum survival kit

Open a file:

```
vim main.c
```

Vim is **modal**. The two modes you need first are:

- **Normal mode** — navigation and commands;
- **Insert mode** — typing text.

Vim starts in Normal mode.

Edit, save, leave

Action	Key / command
Enter Insert mode	**i**
Return to Normal mode	**Esc**
Save	**:w** then **Enter**
Quit	**:q** then **Enter**
Save and quit	**:wq** then **Enter**
Quit and discard changes	**:q!** then **Enter**

If you are unsure which mode you are in, press **Esc**. Pressing it more than once is harmless.

Useful Normal-mode commands

Action	Key / command
Move	arrow keys or **h j k l**
Beginning / end of line	**0** / **$**
First / last line	**gg** / **G**
Delete current line	**dd**
Copy current line	**yy**
Paste after cursor	**p**
Delete one character	**x**
Undo	**u**
Redo	**Ctrl-R**
Search forward	**/text**, then **Enter**
Next / previous match	**n** / **N**
Show line numbers	**:set number**, then **Enter**

Vim emergency exits

If Vim appears to be doing something unexpected:

```
Esc
```

Then choose one:

```
:wq    save and leave
:q!    leave without saving
```

and press **Enter**.

If typed letters are becoming commands instead of text, press **i** to enter Insert mode.

If commands such as `:wq` are becoming part of the file, press **Esc** first.

Editing BVS1 source

Typical workflow:

```
cd ~/teaching-bvs1-code
nano main.c
```

or:

```
cd ~/teaching-bvs1-code
vim main.c
```

After saving, inspect what changed before rebuilding:

```
git diff
```

Then build or run the experiment requested by the current lecture, exercise, or checkpoint.

Remember

- Pick **one** editor and become comfortable with its save/quit workflow.
- Save before building.
- Use `git diff` to check what you actually changed.
- In Vim, **Esc** gets you back to Normal mode.
- In Nano, the command hints at the bottom of the screen are there to help.